A B C A L C U L U S

CHAPTER 8

Applications of Integration

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **8.1 – 8.12** AB

Name: _____ Period: _____

PART I

Sections 8.1 – 8.8

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 8.1 Average Value No Calculator

The average value of $f(x) = x^2$ on $[0, 3]$ is

(A) 1 **(B)** 2

(C) 3 **(D)** 9

Q2 EASY 8.2 Motion (Displacement) No Calculator

A particle moves along the x-axis with velocity $v(t) = t - 2$ for $0 \le t \le 3$. The displacement of the particle on this interval is

(A) $-\frac{3}{2}$ **(B)** $-\frac{1}{2}$

(C) $\frac{1}{2}$ **(D)** $\frac{3}{2}$

Q3 MEDIUM 8.2 Total Distance No Calculator

A particle moves with velocity $v(t) = t - 2$ for $0 \le t \le 3$. The total distance traveled is

(A) $\frac{3}{2}$ **(B)** 2

(C) $\frac{5}{2}$ **(D)** 3

Q4 MEDIUM 8.3 Accumulation Calculator

Water flows into a tank at a rate of $r(t) = 4 + \sin(t)$ gallons per minute, for $0 \le t \le 6$. The tank initially contains 50 gallons. The amount in the tank at $t = 6$ is closest to

(A) 24 gal

(B) 74 gal

(C) 76 gal

(D) 79 gal

Q5 EASY 8.4 Area (in x) No Calculator

The area enclosed by $y = x$ and $y = x^2$ is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) 1

Q6 MEDIUM 8.5 Area (in y) No Calculator

Which integral gives the area enclosed by $x = y^2$ and $x = y + 2$?

(A) $\int_{-1}^{2} \left[(y+2) - y^2\right] dy$

(B) $\int_{-1}^{2} \left[y^2 - (y+2)\right] dy$

(C) $\int_{0}^{4} \left[\sqrt{x} - (x-2)\right] dx$

(D) $\int_{0}^{4} \left[(x-2) - \sqrt{x}\right] dx$

Q7 HARD 8.6 Multiple Intersections No Calculator

The total area of the regions enclosed between $y = x^3 - x$ and the x-axis is

(A) 0

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1

Q8 MEDIUM 8.7 Cross Sections (Squares) No Calculator

The base of a solid is the region in the xy-plane bounded by $y = x$, $y = 0$, and $x = 2$. Cross sections perpendicular to the x-axis are squares. The volume of the solid is

(A) $\frac{4}{3}$

(B) $\frac{8}{3}$

(C) 4

(D) 8

Q9 HARD 8.8 Cross Sections (Triangles) No Calculator

The base of a solid is the region bounded by $y = \sqrt{x}$ and the x-axis on $[0, 4]$. Cross sections perpendicular to the x-axis are equilateral triangles. The volume is

(A) $2\sqrt{3}$

(B) $\frac{4\sqrt{3}}{3}$

(C) $\sqrt{3}$

(D) $\frac{8\sqrt{3}}{3}$

Q10 EASY 8.9 Disc Method No Calculator

The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is revolved about the x-axis. The volume is

(A) 4π

(B) 8π

(C) 16π

(D) $\frac{32\pi}{3}$

Q11 MEDIUM 8.10 Disc (Other Axes) No Calculator

The region bounded by $y = x^2$, $y = 0$, and $x = 2$ is revolved about the line $y = -1$. Which integral gives the volume?

(A) $\pi \int_0^2 (x^2)^2 \, dx$

(B) $\pi \int_0^2 (x^2 + 1)^2 \, dx$

(C) $\pi \int_0^2 \left[(x^2 + 1)^2 - 1\right] dx$

(D) $\pi \int_0^2 (x^2 - 1)^2 \, dx$

Q12 HARD 8.11 Washer (about x-axis) No Calculator

Let R be the region enclosed by $y = x$ and $y = x^2$. The volume of the solid formed when R is revolved about the x-axis is

(A) $\frac{2\pi}{15}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{15}$

(D) $\frac{\pi}{30}$

Q13 HARD 8.12 Washer (Other Axes) No Calculator

Let R be the region bounded by $y = x^2$ and $y = 4$. Which integral gives the volume of the solid generated when R is revolved about the line $y = 5$?

(A) $\pi \int_{-2}^{2} \left[(5 - x^2)^2 - 1\right] dx$

(B) $\pi \int_{-2}^{2} \left[1 - (5 - x^2)^2\right] dx$

(C) $\pi \int_{-2}^{2} (5 - x^2)^2 \, dx$

(D) $\pi \int_{-2}^{2} \left[(x^2 - 5)^2 - (4 - 5)^2\right] dx$

Q14 MEDIUM 8.3 Tabular Accumulation No Calculator

The rate at which people enter a park is modeled by $E(t)$ people per hour, where t is hours since opening. Selected values:

t (hr)	0	2	4	6	8
$E(t)$	100	240	380	300	150

Using a left Riemann sum with the four subintervals of equal length, the approximate total number of people who entered during the 8 hours is

(A) 1340

(B) 2040

(C) 2440

(D) 2640

Q15 EASY 8.1 Average Value No Calculator

If $f(x) = 4x$, the average value of f on $[1, 3]$ is

(A) 4

(B) 6

(C) 8

(D) 12

Q16 MEDIUM 8.2 Position from Velocity No Calculator

A particle has velocity $v(t) = 3t^2 - 6t$ and initial position $x(0) = 2$. Then $x(2) =$

(A) -2

(B) 0

(C) 2

(D) 4

Q17 MEDIUM 8.4 Area between Curves No Calculator

The area enclosed by $y = 4 - x^2$ and the x-axis is

(A) $\frac{8}{3}$

(B) $\frac{16}{3}$

(C) $\frac{32}{3}$

(D) 16

Q18 HARD 8.7 Cross Sections (Semicircles) No Calculator

The base of a solid is the region under $y = \sqrt{x}$ on $[0, 4]$. Cross sections perpendicular to the x-axis are semicircles with diameter in the base. The volume is

(A) $\frac{\pi}{2}$

(B) π

(C) 2π

(D) 4π

PART II *Show All Work*

Free-Response Questions

Free-response answers must include setup, units, and interpretation in context where appropriate. A calculator is permitted unless marked otherwise.

FRQ 1 EASY 8.1 / 8.2 Motion No Calculator

A particle moves along the x-axis with velocity $v(t) = t^2 - 4t + 3$ for $0 \leq t \leq 4$. The particle is at position $x = 2$ when $t = 0$.

(a) Find the displacement of the particle on $[0, 4]$.

(b) Find the position of the particle at $t = 4$.

(c) Find the average value of the velocity on $[0, 4]$.

(d) Find the total distance traveled on $[0, 4]$.

FRQ 2 MEDIUM 8.4 / 8.7 / 8.9 Area & Volume Calculator

Let R be the region in the first quadrant bounded by the graphs of $y = \sin(\pi x)$ and $y = x - x^2$.

(a) Find the area of the region R.

(b) Find the volume of the solid generated when R is revolved about the x-axis.

(c) The region R is the base of a solid. For each x, the cross section perpendicular to the x-axis is a square. Write, but do not evaluate, an integral expression for the volume.

FRQ 3 MEDIUM 8.3 Accumulation (Applied) Calculator

Water is being pumped into a tank at a rate of $P(t) = 20 + 5\sin(t/2)$ gallons per minute. At the same time, water is leaking out at a rate of $L(t) = 2 + 0.5t$ gallons per minute, for $0 \leq t \leq 30$. At time $t = 0$, the tank contains 400 gallons of water.

(a) How many gallons of water are pumped into the tank during the first 30 minutes? Show the setup for your integral.

(b) Write an expression, involving an integral, for the amount of water in the tank at time t, for $0 \leq t \leq 30$.

(c) Find the amount of water in the tank at $t = 30$. Indicate units of measure.

(d) Is the amount of water in the tank increasing or decreasing at $t = 15$? Justify.

FRQ 4 HARD 8.4 / 8.8 / 8.11 / 8.12 Volume Setup Calculator

Let R be the region enclosed by the graphs of $y = e^{-x^2}$ and $y = \frac{1}{2}$.

(a) Find the area of R.

(b) Set up, but do not evaluate, an integral expression for the volume of the solid generated when R is revolved about the x-axis.

(c) Set up, but do not evaluate, an integral expression for the volume of the solid generated when R is revolved about the line $y = -1$.

(d) R is the base of a solid whose cross sections perpendicular to the x-axis are isosceles right triangles with one leg in the base. Write, but do not evaluate, an integral expression for the volume.

FRQ 5 HARD 8.2 / 8.3 Table-Based No Calculator

A car travels along a straight road for 12 seconds. The car's velocity $v(t)$, in meters per second, is differentiable. Selected values are given.

t (sec)	0	3	6	9	12
$v(t)$ (m/s)	0	12	20	15	6

(a) Using a midpoint Riemann sum with two equal subintervals of length 6, approximate $\int_0^{12} v(t)\,dt$. Using correct units, explain the meaning of this integral in context.

(b) Using a trapezoidal sum with the four subintervals shown, approximate $\frac{1}{12}\int_0^{12} v(t)\,dt$. Using correct units, explain the meaning of this value in context.

(c) Must there exist a time t, $3 < t < 9$, at which the acceleration of the car is zero? Justify your answer.

(d) For $0 \le t \le 12$, suppose the car's acceleration is $a(t) = v'(t)$. Is the trapezoidal approximation in part (b) an over- or underestimate of the true average velocity if v is concave down on $(0, 12)$? Justify.